

# SOLUTIONS

1.  $1/x$  is a rational function that has a non-removable discontinuity at  $x = 0$ . Moreover, the left-hand limit of  $f(x)$  approaches 1, while the right-hand limit approaches 0. Since  $\lim_{x \rightarrow 1^-} f(x) = 1$ ,  $\lim_{x \rightarrow 1^+} f(x) = 0$ , the left and right-hand side limits disagree, and therefore we have a jump discontinuity at  $x = 1$ .
2. (a)  $f(a) = \frac{1}{-1} = -1$ ,  $f(b) = \frac{1}{1} = 1$ , so  $[f(a), f(b)] = [-1, 1]$ .  
 (b) Lowkey other than  $-1$  and  $1$ ,  $f(x)$  doesn't obtain any value within that interval. So let's say  $f(x)$  doesn't obtain 0.  
 (c) The IVT states that for a *continuous function* over  $[a, b]$ , there must exist some  $c \in [a, b]$  such that  $f(c) \in [f(a), f(b)]$ . However,  $f(x)$  is not continuous over the stated interval.
3. We set up the bounded equation

$$\begin{aligned} \mathcal{W}_0 - \varepsilon &\leq \mathcal{W} \leq \mathcal{W}_0 + \varepsilon \\ \mathcal{W}_0 - \varepsilon &\leq C_1 v^2 \leq \mathcal{W}_0 + \varepsilon \\ \frac{\mathcal{W}_0 - \varepsilon}{C_1} &\leq v^2 \leq \frac{\mathcal{W}_0 + \varepsilon}{C_1} \\ \sqrt{\frac{\mathcal{W}_0 - \varepsilon}{C_1}} &\leq v \leq \sqrt{\frac{\mathcal{W}_0 + \varepsilon}{C_1}}. \end{aligned}$$

Depending on values we choose for  $\mathcal{W}_0, \varepsilon, C_1$ , we can choose the expression that minimizes the difference between itself and  $v$ .

4. (a) We have

$$(x + h)^n = \sum_{k=0}^n \binom{n}{k} x^{n-k} h^k = x^n + nx^{n-1}h + \mathcal{O}(h^2), \quad (1)$$

where  $\mathcal{O}(h^2)$  bundles up all terms that have  $h^2$  in it or higher powers of  $h$ .

- (b) The numerator, simplified, is equal to the expression  $nx^{n-1}h + \mathcal{O}(h^2)$ .
- (c), (d) The limit, once simplified by canceling with the denominator, is

$$L = \lim_{h \rightarrow 0} (nx^{n-1} + \mathcal{O}(h)). \quad (2)$$

Once we take the limit, everything in  $\mathcal{O}(h)$  vanishes, leaving us with the desired equation that  $dx^n/dx = nx^{n-1}$ .

5. (a) We see that

$$\begin{aligned}
 L &= \frac{d \ln(x)}{dx} = \lim_{h \rightarrow 0} \frac{\ln(x+h) - \ln(x)}{h} \\
 &= \lim_{h \rightarrow 0} \ln \left( \left( \frac{x+h}{x} \right)^{1/h} \right) \\
 \implies e^L &= \lim_{h \rightarrow 0} \left( \frac{x+h}{x} \right)^{1/h} \\
 &= \lim_{h \rightarrow 0} \left( 1 + \frac{1/x}{1/h} \right)^{1/h} \\
 &= \lim_{\eta \rightarrow \infty} \left( 1 + \frac{1/x}{\eta} \right)^{\eta} \\
 &= e^{1/x}.
 \end{aligned}$$

Therefore  $L = 1/x$ .

(b) We utilize some trigonometric identities:

$$\begin{aligned}
 \frac{d \sin(x)}{dx} &= \lim_{h \rightarrow 0} \frac{\sin(x+h) - \sin(x)}{h} \\
 &= \lim_{h \rightarrow 0} \frac{\sin(x) \cos(h) + \sin(h) \cos(x) - \sin(x)}{h} \\
 &= \lim_{h \rightarrow 0} \frac{\sin(h)}{h} \cos(x) + \frac{(\cos(h) - 1)}{h} \sin(x) \\
 &= \cos(x).
 \end{aligned}$$

One can show that the cosine limit vanishes via expanding, to second order, its Taylor series.

(c) Take the hint:

$$\begin{aligned}
 \frac{d}{dx} C^x &= \lim_{h \rightarrow 0} \frac{C^{x+h} - C^x}{h} \\
 &= \lim_{h \rightarrow 0} \frac{C^x (C^h - 1)}{h} \\
 &= C^x \lim_{h \rightarrow 0} \frac{C^h - 1}{h} \\
 &= C^x \lim_{h \rightarrow 0} \frac{1 + (h \ln C) + \mathcal{O}(h^2) - 1}{h} \\
 &= \ln(C) \cdot C^x.
 \end{aligned}$$

6. We know that at time  $t$ , the horizontal velocity,  $dx/dt$ , is  $-2$  meters per second. We also know that  $x^2 + y^2 = 10^2$  must hold true at all times. We can now implicitly differentiate both sides with respect

to  $t$  (assuming  $x, y = x(t), y(t)$ ) to find that

$$2x \frac{dx}{dt} + 2y \frac{dy}{dt} = 0, \quad \text{meaning} \quad x \frac{dx}{dt} = -y \frac{dy}{dt}. \quad (3)$$

We're also given the horizontal and vertical lengths of the ladder at time  $t$ , so we can use that to help us solve  $dy/dt$ . Doing algebra will give you that

$$\frac{dy}{dt} = -\frac{x}{y} \frac{dx}{dt} = -\left(-\frac{8}{6}\right)(-2) = -\frac{8}{3} \text{ meters per second.} \quad (4)$$

Hence, the vertical velocity is decreasing at  $8/3$  meters per second at time  $t$ .

7. (a) Generally,  $V = \frac{1}{3}\pi r^2 h$ , but using the figure and what's given in the question, we can find that  $r(h) = h$ , and therefore  $V = \frac{1}{3}\pi h^3$ .

(b) Implicit differentiation yields

$$\frac{dV}{dt} = \pi h^2 \frac{dh}{dt}. \quad (5)$$

- (c) We know that  $V_{\text{pool}} = \frac{1}{3}\pi(10)^3 = \frac{1000}{3}\pi$  cubic meters, so we solve  $\frac{500}{3}\pi = \frac{1}{3}\pi h^3 \implies h = \sqrt[3]{500}$  meters. With knowledge that  $dV/dt = 20$  liters per minute, or  $0.02$  cubic meters per minute, we have, using the preceding equation,

$$0.02 = \pi \cdot 500^{2/3} \cdot \frac{dh}{dt} \implies \frac{dh}{dt} \approx 1.01 \cdot 10^{-4} \text{ m/min.} \quad (6)$$

8. (a) As the hint suggests, we will use the mean value theorem. Observe that  $\sigma(-\ln 2) = 1/3$ , while  $\sigma(\ln 2) = 2/3$ , and therefore we are guaranteed that  $\sigma'(x)$  must obtain  $\sigma'(c) = (2/3 - 1/3)/(2 \ln 2) = 1/\ln 64$  for some  $c \in [-\ln 2, \ln 2]$ .

(b) Take the derivative of  $\sigma(x)$  by utilizing the chain rule once to obtain

$$\sigma'(x) = \frac{e^{-x}}{(1 + e^{-x})^2} = \frac{1}{2(1 + \cosh x)}, \quad \text{where } \cosh x = \frac{e^x + e^{-x}}{2}. \quad (7)$$

We then find  $c$  via:

$$\begin{aligned} \frac{1}{\ln 64} &= \frac{1}{2(1 + \cosh c)} \\ \implies c &= \cosh^{-1} \left( \frac{1}{2} \ln(64) - 1 \right) \\ &\approx 0.396. \end{aligned}$$

*Note:  $c \approx -0.396$  is also a valid solution since  $\cosh$  is an even function.*

We proceed to now find the maximum of  $\sigma'(x)$  over all real numbers, and we may do so by

evaluating  $\sigma''(x)$  and setting it to 0:

$$\begin{aligned}
\sigma''(x) &= \frac{d}{dx} \frac{e^{-x}}{(1+e^{-x})^2} \\
&= -\frac{e^{-x}}{(1+e^{-x})^2} + \frac{2e^{-2x}}{(1+e^{-x})^3} \\
\therefore 0 &= -\frac{e^{-x}}{(1+e^{-x})^2} + \frac{2e^{-2x}}{(1+e^{-x})^3} \\
&= 2e^{-x} - (1+e^{-x}) \\
&\implies x = 0 \\
\therefore \sigma'(0) &= \frac{1}{4}.
\end{aligned}$$

- (c) First, we see that directly taking the limit  $\lim_{x \rightarrow -\infty} \sigma'(x)$  yields the indeterminate expression  $\infty/\infty$ , and this permits us to use l'Hopital's rule (the rule may also be applied when direct evaluation of the limit produces  $0/0$ ). Taking the derivative of the numerator and denominator and simplifying the resulting quotient, we have the expression  $1/(2(1+e^{-x}))$ , which, upon another direct evaluation of the limit, immediately vanishes.
- (d) From (b), we observe that in solving for when  $\sigma''(x) = 0$ , we eventually had to solve  $e^{-x} = 1$ , which uniquely admits  $x = 0$  as a solution. One can check that on either side of  $x = 0$ ,  $\sigma''(x)$  has opposing signs, implying  $x = 0$  is an inflection point.
- (e) There are a few ways to show this. Here we present one way. Rewrite  $\sigma''(x)$  to obtain

$$\sigma''(x) = \frac{e^{-x}(e^{-x} - 1)}{(1+e^{-x})^3}. \quad (8)$$

When  $x > 0$ ,  $e^{-x} > 0$ ,  $(1+e^{-x})^3 > 0$ , and  $e^{-x} - 1 < 0$ , which ensures that  $\sigma''(x) < 0$  for all positive  $x$ . Since this condition implies a concave down function, we have that  $\sigma(x)$  is concave down over  $(0, \infty)$ .

- (f) We use our expression for  $\sigma'(x)$  earlier and see whether we can construct it out of some combination of  $\sigma(x)$ . While not initially obvious, we can do this via partial fraction decomposition, making an appropriate substitution  $e^{-x} \mapsto u$  for convenience:

$$\begin{aligned}
\text{Let } \frac{u}{(1+u)^2} &= \frac{A}{1+u} + \frac{B}{(1+u)^2} \\
&= \frac{A(1+u) + B}{(1+u)^2} \\
&= \frac{(A+B) + Au}{(1+u)^2}.
\end{aligned}$$

Therefore, matching coefficients on both sides of the equation,  $A = -B$ , while  $A = 1$ , and therefore we see that

$$\frac{u}{(1+u)^2} = \frac{1}{1+u} - \frac{1}{(1+u)^2} = \sigma(x) - \sigma^2(x) = \sigma(1-\sigma). \quad (9)$$

Consequently, the differential equation for which  $\sigma$  is a solution is  $\sigma' = \sigma(1-\sigma)$ .

(g) Recall that our differential equation is  $\sigma' = \sigma(1 - \sigma)$ .

- i. When  $\sigma \ll 1$ ,  $\sigma'$  is also very small, and this represents a sparse population struggling to grow or at the very beginning stages of exponential growth. For instance, one bacteria with a doubling time of  $\tau$  will take  $2\tau$  units of time to quadruple, or increase its population by three bacteria. However, from a much larger population, say 1000 bacteria, it would only take less than  $\tau/100$  units of time for its population to increase by three.
- ii. When  $\sigma \sim 1/2$ ,  $\sigma'$  is maximized, and growth is also therefore maximized. This is the period most resembling what people may think of as “exponential growth”; the population is consuming environmental resources and growing maximally.
- iii. When  $\sigma$  is close to 1, the population has nearly reached its carrying capacity, and its behavior is like that when  $\sigma \ll 1$  — growth slows due to competition over resources and possibly overcrowding. At the population’s carrying capacity ( $\sigma = 1$ ), the net birth and death rates of members in the population are equal.
- iv. Finally, when  $\sigma > 1$ , observe that  $\sigma'$  is negative, and therefore the population must shrink. This makes sense, since having more members of the population than the environment can sustain means some members must die. As  $\sigma$  becomes larger, say for an extremely large initial starting population,  $\sigma'$  proportionally becomes more negative.

9. First, we’ll begin with a quick derivation of the arc length formula. The arc length ( $s$ ) for any integrable function  $y = f(x)$  over a sufficiently small interval  $\delta x$  is approximately  $\delta s^2 \sim \delta x^2 + \delta y^2$ , as per the Pythagorean theorem. As such,  $ds = \sqrt{dx^2 + dy^2}$ , and therefore over a curve  $\gamma$ ,

$$s = \int_{\gamma} ds = \int_{\gamma} \sqrt{dx^2 + dy^2} = \int_a^b \sqrt{1 + (y')^2} dx. \quad (10)$$

Here,  $y = \sin x$ , and therefore  $y' = \cos x$ , so the arc length as described in the problem is

$$s = \int_0^{2\pi} \sqrt{1 + \cos^2(x)} dx \approx 7.64. \quad (11)$$

10. (a)  $I_1$  is best handled by  $u$ -substitution via  $u = r^2$ , meaning  $du = 2r dr$  and  $\{0, \infty\}$ , the bounds of integration, become  $\{0^2, \infty^2\} = \{0, \infty\}$ . Ultimately, this gives us

$$I_1 \mapsto \frac{1}{2} \int_0^{\infty} e^{-u} du = \frac{1}{2}. \quad (12)$$

- (b) We proceed via integration by parts, where  $u = x$ ,  $dv = \sin(x) dx$ , and also because  $x \sin(x)$  is even, we have

$$I_2 = 2 \int_0^{\pi} \underbrace{x}_u \underbrace{\sin(x) dx}_{dv}. \quad (13)$$

Since  $\int u dv = uv - \int v du$ , and we have  $du = dx$ ,  $v = -\cos(x)$ , we have everything we need:

$$I_2 = 2 \left( \underbrace{-x \cos(x)}_{\pi} \Big|_0^{\pi} + \underbrace{\int_0^{\pi} \cos(x) dx}_0 \right) \quad (14)$$

$$= 2\pi. \quad (15)$$

- (c) The integrand  $\vartheta(t) = f(t) - f(-t)$  is an odd function, since  $\vartheta(-t) = f(-t) - f(t) = -(f(t) - f(-t)) = -\vartheta(t)$ . Therefore, since we integrate over symmetric bounds from  $-a$  to  $a$ ,  $I_3 = 0$ .
- (d) While it might be possible to proceed via integration by parts, one may get stuck in loops (I forgot if this integral suffers from that problem) where integration by parts is carried out *ad infinitum*. Instead, we can employ a double angle identity, particularly that  $\sin^2(x) = \frac{1}{2}(1 - \cos(2x))$ . Then we can integrate in a relatively straightforward manner:

$$I_4 = \frac{1}{2} \int_0^\pi (1 - \cos(2x)) dx \quad (16)$$

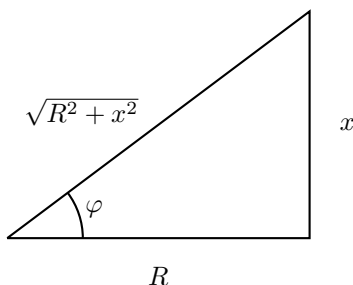
$$= \frac{1}{2} \left( \pi - \frac{1}{2} \sin(2x) \right) \Big|_0^\pi \quad (17)$$

$$= \frac{\pi}{2}. \quad (18)$$

- (e) This integral can be tackled using a trigonometric  $u$ -sub. First, we factor out the constant in the denominator, giving us the integral

$$\int_{-a}^a (R^2 + x^2)^{-3/2} dx \quad \mapsto \quad \frac{1}{R^3} \int_{-a}^a \left( 1 + \left( \frac{x}{R} \right)^2 \right)^{-3/2} dx. \quad (19)$$

Now suppose we draw a right triangle and label the edges accordingly:



We see that if we substitute  $x \mapsto \varphi$ , we can write  $\tan \varphi = x/R$ , and taking derivatives of both sides gives  $\sec^2 \varphi d\varphi = dx/R$ , which we substitute into the differential  $dx$ . Now, since  $1 + \tan^2 \varphi = \sec^2 \varphi$ , we substitute this into the integrand as well, and after some algebra, we have

$$I_5 = \frac{1}{R^2} \int_I \cos \varphi d\varphi, \quad (20)$$

where  $I$  is the relevant interval to integrate over after performing the substitution. We won't bother actually calculating what the bounds of  $I$  are, since first of all, it's pretty messy, and second of all, we can just plug in our original expression after integration and evaluate at  $\{-a, a\}$ . Integrating  $\cos \varphi$  is relatively straightforward; it yields  $\sin \varphi$ . Looking back at our triangle, we recognize that

$$\sin \varphi = \frac{x}{\sqrt{R^2 + x^2}}, \quad (21)$$

and we arrive at our final answer:

$$I_5 = \frac{1}{R^2} \frac{x}{\sqrt{R^2 + x^2}} \Big|_{-a}^a = \boxed{\frac{2a}{R^2 \sqrt{R^2 + a^2}}}. \quad (22)$$

11. Set the vertex of the cone at  $x = 0$ , meaning that at  $x = h$  we have  $f(h) = r$ ; a line with slope  $r/h$ . We can revolve this around the  $x$  axis to obtain the volume of the resulting solid. This is done via the disk method, where the volume  $V$  can be obtained using

$$V = \int_0^h \pi f(x)^2 dx. \quad (23)$$

With  $f(x) = rx/h$ , we can substitute directly to get

$$V = \int_0^h \pi (rx/h)^2 dx = \frac{\pi r^2}{h^2} \cdot \frac{x^3}{3} \Big|_0^h = \frac{1}{3} \pi r^2 h. \quad (24)$$

12. Let's first derive the area formula in polar coordinates. If we consider an arbitrarily small angular displacement  $d\theta$ ,  $f(\theta)$  remains roughly constant (to first order) and sweeps out an arc length that is about  $f(\theta) d\theta$  (again, to first order). We may treat this as an isosceles triangle with  $f(\theta)$  making up the two equal-length sides and  $f(\theta) d\theta$  making up the base. Computing the area gives  $dA = \frac{1}{2} f(\theta)^2 d\theta$ , and following this logic, we have

$$A = \frac{1}{2} \int_{\theta_1}^{\theta_2} f(\theta)^2 d\theta. \quad (25)$$

What remains now is to find  $\theta_1, \theta_2$  which bounds one of the "petals" mentioned in the question. We can do this by solving for  $r = 0$  over  $[-\pi, \pi]$ , which gives us  $\theta = -\frac{3\pi}{4}, -\frac{\pi}{4}, \frac{\pi}{4}, \frac{3\pi}{4}$ . Let's choose  $\theta_1, \theta_2 = -\frac{\pi}{4}, \frac{\pi}{4}$  as our bounds of integration, and we now proceed to find the area of our petal,  $A$ :

$$A = \int_{-\pi/4}^{\pi/4} \frac{1}{2} (\cos 2\theta)^2 d\theta \quad (26)$$

$$= \frac{1}{4} \int_{-\pi/4}^{\pi/4} (1 + \cos 4\theta) d\theta \quad (27)$$

$$= \frac{1}{4} \left( \frac{\pi}{2} + \frac{1}{4} \sin 4\theta \Big|_{-\pi/4}^{\pi/4} \right) \quad (28)$$

$$= \frac{\pi}{8}. \quad (29)$$

13. For  $p \geq 2$ , we hope to compare  $S$  against an integral that serves as an upper bound. If this integral converges, then  $S$  must also converge. First, from  $S$ , remove the  $n = 1$  term, which is always equal to 1; we will see why this step is important momentarily. Next, since terms in  $S$  corresponding to the same index are smaller the larger  $p$  gets (for instance,  $\frac{1}{2^3} < \frac{1}{2^2}$ ), if we can show that  $p = 2$  converges, all other  $p > 2$  necessarily converges as well.

Denote  $S - 1$  as  $S$  with the first term removed. We now make another observation:  $S - 1$  for  $p = 2$  is equal to the right-handed Riemann sum of  $x^{-2}$  from 1 to  $\infty$  with unit intervals, and because  $x^{-2}$  is a monotonically decreasing function over the positive real numbers, the right-handed Riemann sum is an underestimate of the true integral. In other words,  $S - 1 \leq \int_1^\infty x^{-2} dx$ . Now, using the power rule,

$$\int_1^\infty x^{-2} dx = \lim_{a \rightarrow \infty} \frac{1}{x} \Big|_a^1 = 1. \quad (30)$$

Therefore,  $S - 1 \leq 1$ , and so  $S \leq 2 < \infty$ , meaning  $S$  converges. As  $p \rightarrow \infty$ , all but the first term vanish, and as such  $S \rightarrow 1$ .

14. The general formula for a Taylor series about  $x_0$  is

$$f(x) = \sum_{k=0}^{\infty} \frac{f^{(k)}(x_0)}{k!} (x - x_0)^k. \quad (31)$$

A Maclaurin series imposes the condition that  $x_0 = 0$ .

(a)  $f_1$  can be interpreted as a geometric series:

$$f_1(x) = 1 + x^2 + x^4 + \cdots = \sum_{k=0}^{\infty} x^{2k}. \quad (32)$$

Use the ratio test to determine radius of convergence, which evaluates to the condition  $|x^2| < 1$ , or equivalently that this series converges within the interval  $(-1, 1)$ .

(b) Since we know that  $\frac{d^4}{dx^4} \sin(x) = \sin(x)$ , we only need to evaluate four derivatives to get all coefficients of the Maclaurin series. We have

$$\sin(0) = 0, \quad \sin'(0) = \cos(0) = 1, \quad \sin''(0) = -\sin(0) = 0, \quad \sin^{(3)}(0) = -\cos(0) = -1. \quad (33)$$

As such, the first three nonzero terms are

$$f_2(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!} + \cdots. \quad (34)$$

We observe that this pattern continues, and using this, write the series expansion of  $f_2$  as a closed-form sum:

$$\sin(x) = \sum_{k=0}^{\infty} \frac{(-1)^k x^{2k+1}}{(2k+1)!}. \quad (35)$$

(c) Unfortunately,  $\tan(x)$  is not cyclic in its derivatives, so we actually have to manually evaluate some derivatives. Note that  $\tan(0) = 0$ , which will help us when we need to evaluate higher-order coefficients. We see that

$$f_3'(x) = \sec^2(x), \quad (36)$$

$$f_3''(x) = 2 \sec^2(x) \tan(x), \quad (37)$$

$$f_3^{(3)}(x) = 2 \sec^4(x) + 4 \tan^2(x) \sec^2(x), \quad (38)$$

$$f_3^{(4)}(x) = 16 \tan(x) \sec^4(x) + 8 \tan^3(x) \sec^2(x), \quad (39)$$

$$f_3^{(5)}(x) = 16 \sec^6(x) + 88 \tan^2(x) \sec^4(x) + 16 \tan^4(x) \sec^2(x). \quad (40)$$

From here we see that  $f_3(0) = f_3''(0) = f_3^{(4)}(0) = 0$ , while  $f_3'(0) = 1, f_3^{(3)}(0) = 2, f_3^{(5)}(0) = 16$ , meaning our first three nonzero terms in the expansion for  $f_3(x)$  are

$$\tan(x) = x + \frac{x^3}{3} + \frac{2x^5}{15} + \cdots. \quad (41)$$



(d) Let's take a few derivatives to see if we can spot a pattern:

$$f_4'(x) = \frac{1}{1+x}, \quad f_4''(x) = -\frac{1}{(1+x)^2}, \quad f_4^{(3)}(x) = \frac{2}{(1+x)^3}, \quad f_4^{(4)}(x) = -\frac{6}{(1+x)^4}, \quad \dots \quad (42)$$

It appears that

$$f_4^{(n)}(x) = (-1)^{n-1} \frac{(n-1)!}{(1+x)^n} \quad \text{for } n \geq 1, \quad (43)$$

and as a result  $f_4^{(n)}(0) = (-1)^{n-1}(n-1)!$ . Substituting this as our coefficient for the Maclaurin series, we get

$$\ln(1+x) = \sum_{k=1}^{\infty} (-1)^{k-1} \frac{x^k}{k} = x - \frac{x^2}{2} + \frac{x^3}{3} + \dots \quad (44)$$

Note that this series also only has a radius of convergence of 1; using the ratio test, we end up with the condition  $|x| < 1$ .

15. We can find the Maclaurin series for  $e^x$ , which is

$$e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!}. \quad (45)$$

The desired identity is immediately derived when we set  $x = 1$ .

16. (a) From 14(b), we know that the first higher-order term for  $\sin(\theta)$  is a cubic term. If we want to find approximately what  $\theta$  causes this cubic term to contribute more than 10% to the linear term (which is  $\theta$ ), we can solve the equation  $\theta/10 = \theta^3/3!$ , or that  $\theta = \pm\sqrt{3/5}$  rads.

(b) We observe that

$$\dot{\theta}\ddot{\theta} = \frac{d}{dt} \left( \frac{1}{2}(\dot{\theta})^2 \right), \quad \dot{\theta} \sin \theta = \frac{d}{dt} (-\cos \theta), \quad (46)$$

and this allows us to group terms together and conclude that

$$\frac{d}{dt} \left( \frac{1}{2}\dot{\theta}^2 - \frac{g}{\ell} \cos \theta \right) = 0 \quad \implies \quad \frac{1}{2}\dot{\theta}^2 - \frac{g}{\ell} \cos \theta = \alpha. \quad (47)$$

(c) If the pendulum is in the conditions as described, then  $\dot{\theta} = 0$  and  $\theta = \theta_0$ . Since the above equation applies regardless of our choice of time, we can deduce that  $\alpha = -\frac{g}{\ell} \cos \theta_0$ .

(d) We can get the desired equation by simply isolating  $\dot{\theta}$ :

$$\frac{1}{2}\dot{\theta}^2 - \frac{g}{\ell} \cos \theta = -\frac{g}{\ell} \cos \theta_0 \quad \implies \quad \dot{\theta} = \sqrt{\frac{2g}{\ell} (\cos \theta - \cos \theta_0)}. \quad (48)$$

(e) While we can expand  $\cos \theta \simeq 1$ , we don't get anything interesting, and the oscillations aren't periodic. This approximation would not work unless  $\theta$  was so small that the pendulum doesn't oscillate, so we expand to the next term:  $\cos \theta \simeq 1 - \theta^2/2$ . With this, we can do some rewriting,

defining  $\alpha$  as we did above,  $\omega^2 = g/\ell$ , and  $\beta^2 = 2(\alpha + \omega^2)$ :

$$\cos \theta \simeq 1 - \frac{\theta^2}{2} \quad (49)$$

$$\Rightarrow \quad \dot{\theta} = \sqrt{\frac{2g}{\ell}(\cos \theta - \cos \theta_0)} \quad (50)$$

$$= \left( \frac{2g}{\ell} \left( 1 - \frac{\theta^2}{2} \right) + 2\alpha \right)^{1/2} \quad (51)$$

$$= (\beta^2 - \omega^2 \theta^2)^{1/2}. \quad (52)$$

This differential equation is separable, so we will separate and integrate both sides. Now, at some time  $t$ , we know that  $\theta$  will be at  $\theta(t)$ , and at  $t = 0, \theta = \theta_0$ , which we now use as bounds for integration:

$$\int_{\theta_0}^{\theta} \frac{d\vartheta}{(\beta^2 - \omega^2 \vartheta^2)^{1/2}} = \int_0^t dt' \quad (53)$$

$$\Rightarrow \quad \int_{\theta_0}^{\theta} \frac{d\vartheta}{\left(1 - (\omega/\beta)^2 \vartheta^2\right)^{1/2}} = \beta t. \quad (54)$$

Here I will make the  $u$ -sub  $\varphi = \frac{\omega}{\beta} \vartheta$ , meaning  $d\varphi = \frac{\omega}{\beta} d\vartheta$ . Since I'm also kind of lazy, I'll write my change of bounds so that  $\varphi_0 = \varphi(\theta_0)$ ,  $\varphi_t = \varphi(\theta)$ , and we'll now complete the integration:

$$\int_{\varphi_0}^{\varphi_t} \frac{d\varphi}{(1 - \varphi^2)^{1/2}} = \omega t \quad (55)$$

$$\Rightarrow \quad \sin^{-1}(\varphi) \Big|_{\varphi_0}^{\varphi_t} = \omega t \quad (56)$$

$$\Rightarrow \quad \sin^{-1}(\varphi_t) = \omega t + \underbrace{\sin^{-1}(\varphi_0)}_{\delta} \quad (57)$$

$$\therefore \quad \theta(t) = \frac{\beta}{\omega} \sin(\omega t + \delta). \quad (58)$$

A quick evaluation shows that  $\delta = \sin^{-1}(\varphi_0) = \sin^{-1}(\frac{\omega}{\beta} \theta_0)$ , and  $\beta/\omega = \sqrt{2}(1 - \cos \theta_0)$ , so our final solution is

$$\theta(t) = A \sin(\omega t + \delta), \quad (59)$$

where  $A, \omega$ , and  $\delta$  are as stated in the problem.

- (f) Using the information provided in the problem, we know that the time it takes the pendulum to swing from  $\theta = 0$  to  $\theta = \theta_0$  is a quarter of the period,  $T/4$ . This informs our bounds of integration, and after separating the equation like we did in (e) and setting up the integral, we retrieve the desired equation.
- (g) We will use the half-angle identity

$$\sin^2\left(\frac{\theta}{2}\right) = \frac{1 - \cos \theta}{2}. \quad (60)$$

With this, we rewrite the stuff within the square root:

$$\cos \theta - \cos \theta_0 = (1 - 2 \sin^2(\theta/2)) - (1 - 2 \sin^2(\theta_0/2)) = 2(\sin^2(\theta_0/2) - \sin^2(\theta/2)). \quad (61)$$

Once things are simplified, we get equation 21 in the problem document. Let's now proceed to the next step. Defining  $k = \sin(\theta_0/2)$  as suggested, our equation becomes

$$T = \frac{2}{\omega} \int_0^{\theta_0} \frac{d\theta}{\sqrt{k^2 - \sin^2(\theta/2)}}. \quad (62)$$

We eventually want our integrand to look like the one in  $K(k)$ ; we can get a little closer by observing  $\sqrt{k^2 - \sin^2(\theta/2)} = k\sqrt{1 - \frac{1}{k^2} \sin^2(\theta/2)}$ . If we can replace  $\frac{1}{k^2} \sin^2(\theta/2)$  with something related to  $\sin^2 \phi$ , this would be good progress. Moreover, we see that the bounds have been altered such that  $\{0, \theta_0\} \mapsto \{0, \pi/2\}$ . So if  $f(\theta) = g(\phi)$  is our substitution, we know  $f(0) = g(0)$ , and furthermore  $f(\theta_0) = g(\pi/2)$ . We can stare at these conditions for a while and try a few things. Let's say — since we want to replace  $\sin^2(\theta/2)$  with  $\sin^2(\phi)$  as mentioned above — that  $\sin(\theta/2)$  is proportional to  $\sin(\phi)$ . For now, we let them be equal, meaning our substitution is  $\sin(\theta/2) = \sin(\phi)$ , but we will make appropriate modifications based on the present conditions. Now  $f(0) = \sin(0) = 0$  and  $g(0) = \sin(0) = 0$ , and so  $f(0) = g(0)$ . Meanwhile,  $f(\theta_0) = k$ , while  $g(\pi/2) = 1$ , so to make them equal,  $f(\theta) = kg(\phi)$ . In other words, the appropriate substitution is

$$k \sin \phi = \sin \frac{\theta}{2} \implies k \cos \phi d\phi = \frac{1}{2} \cos \frac{\theta}{2} d\theta. \quad (63)$$

Using the identity  $\cos^2 \theta = 1 - \sin^2 \theta$ , we replace  $d\theta$ :

$$k \cos \phi d\phi = \frac{1}{2} \cos \frac{\theta}{2} d\theta \quad (64)$$

$$\implies d\theta = \frac{2k(1 - \sin^2 \phi)^{1/2}}{(1 - \sin^2(\theta/2))^{1/2}} d\phi \quad (65)$$

$$= \frac{2k(1 - \sin^2 \phi)^{1/2}}{(1 - k^2 \sin^2 \phi)^{1/2}} d\phi \quad (66)$$

$$\implies \frac{d\theta}{(k^2 - \sin^2(\theta/2))^{1/2}} = \frac{2k(1 - \sin^2 \phi)^{1/2}}{(1 - k^2 \sin^2 \phi)^{1/2}} \cdot \frac{d\phi}{(k^2 - \sin^2(\theta/2))^{1/2}} \quad (67)$$

$$= \frac{2k(1 - \sin^2 \phi)^{1/2}}{(1 - k^2 \sin^2 \phi)^{1/2}} \cdot \frac{d\phi}{\cancel{k(1 - \sin^2 \phi)^{1/2}}} \quad (68)$$

$$= \frac{2}{(1 - k^2 \sin^2 \phi)^{1/2}} d\phi. \quad (69)$$

Since the left hand side of equation 67 is the original integrand, we replace it with the new integrand and bounds, giving us equation (22) in the problems.

- (h) We essentially need to find the Taylor series of  $(1 - k^2 \sin^2 \phi)^{-1/2}$  up to third nonzero order. To help simplify things, we can introduce the substitution  $x = k^2 \sin^2 \phi$ , and now our task is to find the series expansion of  $(1 - x)^{-1/2}$ , which we can do so about  $x = 0$  to obtain

$$(1 - k^2 \sin^2 \phi)^{-1/2} \simeq 1 + \frac{1}{2} k^2 \sin^2 \phi + \frac{3}{8} (k^2 \sin^2 \phi)^2. \quad (70)$$

While we could be a little more careful and instead use the substitution  $x = k \sin \phi$ , derivatives with odd powers of  $\sin \phi$  vanish. Upon integration, we get

$$I = \int_0^{\pi/2} (1 - k^2 \sin^2 \phi)^{-1/2} d\phi \simeq \frac{\pi}{2} + \frac{\pi}{8} k^2 + \frac{9\pi}{128} k^4. \quad (71)$$

Expand  $k$  in the quadratic term to second order and keep only the first term in the quartic term, so that we end up with

$$I \simeq \frac{\pi}{2} + \frac{\pi}{8} \left( \frac{\theta_0}{2} - \frac{\theta_0^3}{48} \right)^2 + \frac{9\pi}{128} \left( \frac{\theta_0}{2} \right)^4 \quad (72)$$

$$= \frac{\pi}{2} + \frac{\pi}{8} \left( \frac{\theta_0^2}{4} - \frac{\theta_0^4}{48} + \frac{\theta_0^6}{48^2} \right) + \frac{9\pi}{2048} \theta_0^4 \quad (73)$$

$$\simeq \frac{\pi}{2} + \frac{\pi}{32} \theta_0^2 + \frac{11\pi}{6144} \theta_0^4. \quad (74)$$

Combine this back into the period formula to obtain

$$T \simeq \frac{4}{\omega} \left( \frac{\pi}{2} + \frac{\pi}{32} \theta_0^2 + \frac{11\pi}{6144} \theta_0^4 \right) \quad (75)$$

$$= 2\pi \sqrt{\frac{\ell}{g}} \left( 1 + \frac{1}{16} \theta_0^2 + \frac{11}{3072} \theta_0^4 \right). \quad (76)$$

Indeed, the familiar  $T = 2\pi\sqrt{\ell/g}$  appears as the leading order term in this expansion, while higher order contributions come in powers of  $\theta_0^2$ .

When  $\theta_0 = 1^\circ$ , observe that the term in the expansion is (converting to radians)  $(1 + \frac{1}{16}(\pi/180)^2 + \frac{11}{3072}(\pi/180)^4)$ , which corresponds to approximately a 0.002% fractional deviation, so the regular period formula works well here. At  $\theta_0 = 30^\circ$ , this becomes a 1.74% deviation, and at  $\theta_0 = 90^\circ$ , this becomes a 17.6% deviation, which is definitely significant.